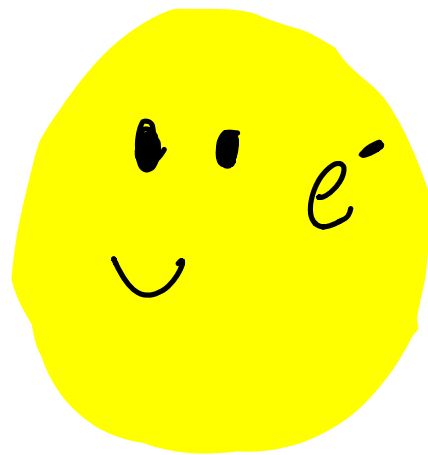
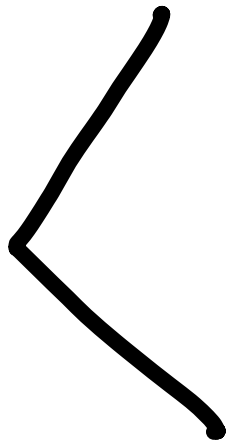


Electroweak Baryogenesis

Q: Why is there more
Matter than Anti-Matter
in our universe?



Cosmology experiments Verity

Asymmetry:

$$\Delta B = 6 \times 10^{-9} n_\gamma$$

"Baryogenesis through String Quarks"

In Cosmic rays: $\frac{P}{\rho} \sim 10^{-4}$

$$\eta = \frac{\Delta n_B}{s} = \frac{n_b - n_{\bar{b}}}{s} \quad \left. \vphantom{\frac{\Delta n_B}{s}} \right\} \frac{\text{baryon}}{\text{entropy}}$$

$$1.5 \times 10^{-10} < \eta < 7 \times 10^{-10}$$

If the Universe originally had this asymmetry it would be washed away by inflation.

Can talk later

Ex:

$$n_B > \bar{n}_B$$

Inflation

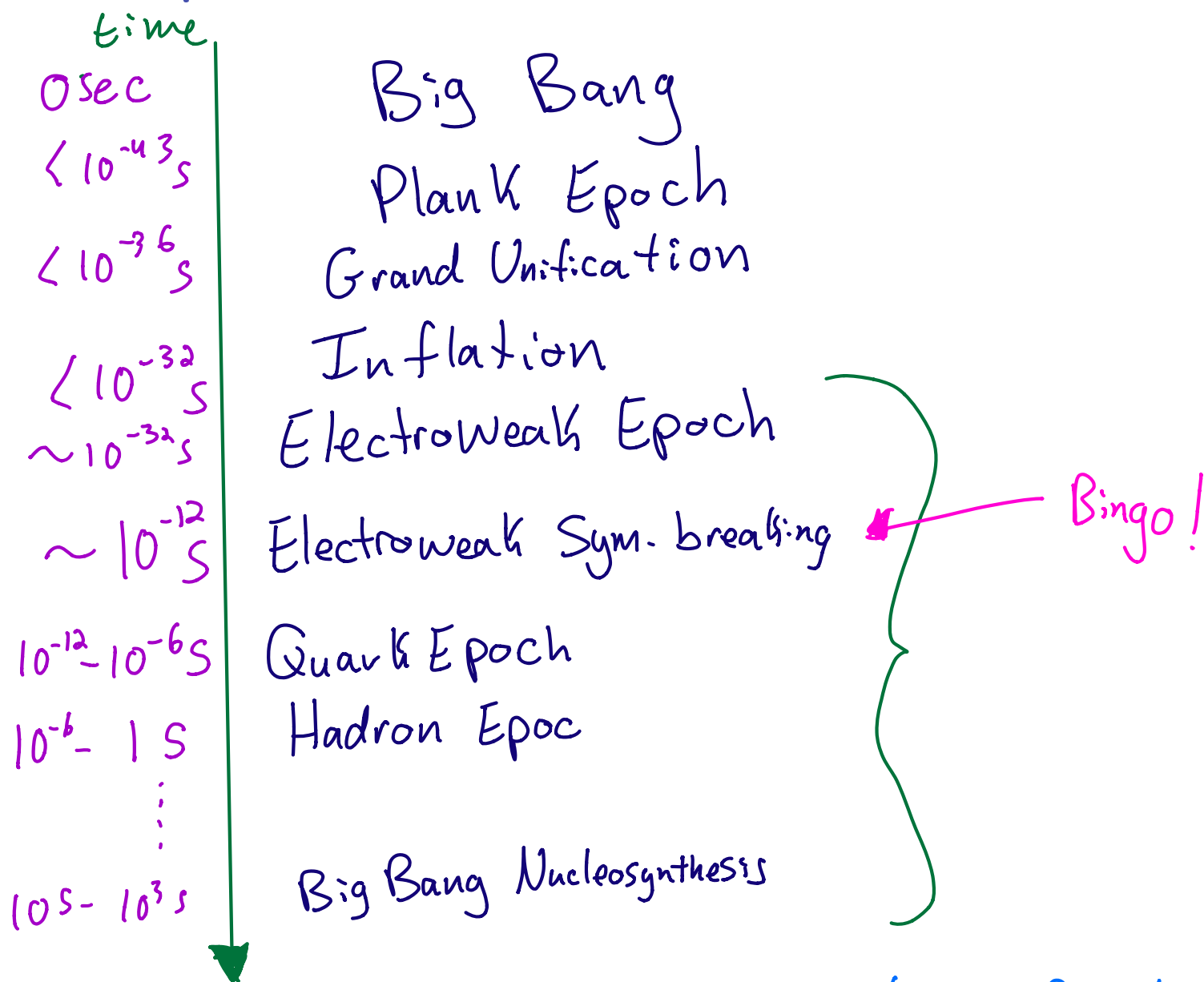
$$\frac{n_B}{V}, \frac{\bar{n}_B}{V} \ll 1$$

reheating

$$< \frac{n_B}{V}, \frac{\bar{n}_B}{V}$$

$$n_B \approx \bar{n}_B$$

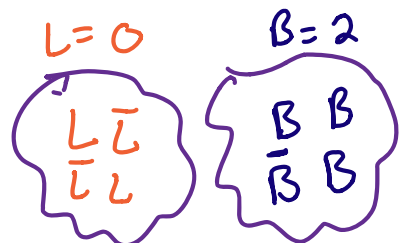
There must be some Mechanism
or process explaining the asymmetry



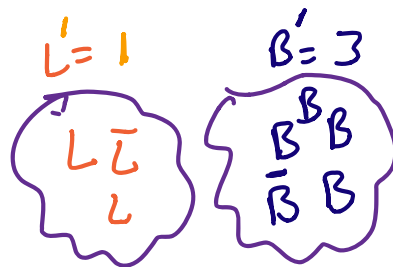
We saw a chiral anomaly for Baryon/Lepton Currents.

$$2n \gamma^{\mu} B \approx F_{\mu\nu} F^{\mu\nu}$$

However taking the difference between both
 $B-L$ would NOT be broken by chiral anomaly



$$\bar{L} \rightarrow B \Rightarrow B-L=2 \quad B'-L'=2$$



Sakharov's 3 Conditions

- Baryon # violation
- C & CP violation
 - ↑ obvious
 - ↑ So that Baryons won't be counter balanced
 - ↑ So that LH Baryons $\leftrightarrow$ RH Anti-Baryons
- No thermal Equilibrium Int.
 - ↑ No CPT business happening

The electroweak phase transition is an incredible place for this to occur.

Any process taking place during this time is called Electroweak Baryogenesis

Recall EWPT

AKA when the HIGGS breaks Symmetry.

$$SU(2) \times U(1) \rightarrow \begin{cases} SU(2) & U(1) \\ \text{4 massless bosons} & \begin{cases} W_1 = B_1 + iB_2 & \gamma = A + iB_3 \leftarrow \text{massless} \\ W_2 = B_2 - iB_1 \\ Z = A - iB \end{cases} \end{cases}$$

With the Gauge + Higgs Lagrangian :

$$\mathcal{L} = (D_\mu \phi)^\dagger (D_\mu \phi) - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{4} W_{\mu\nu}^a W^{a\mu\nu} + V(\phi)$$

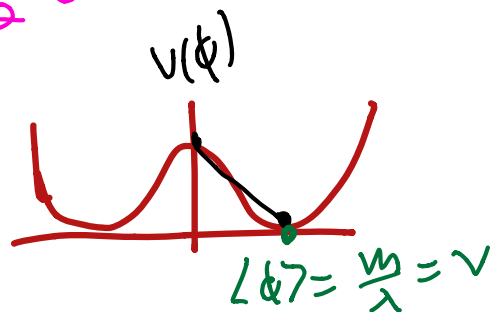
$$V(\phi) = -\frac{m}{2} \phi \phi^\dagger + \frac{1}{4} (\phi \phi^\dagger)^2 = \frac{1}{4} (\phi \phi^\dagger - v)^2 \quad v = \frac{m}{\lambda}$$

$$D_\mu = \partial_\mu - ig Z B_\mu - \frac{i}{2} g' A_\mu$$

$$\theta_w = \arctan\left(\frac{g'}{g}\right)$$

want $\theta_w \approx 0$

Tree Order :



Not Quiet the whole story!

Temperature QFT

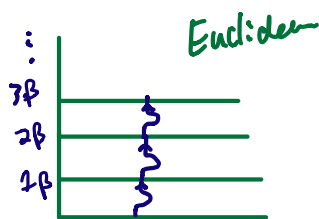
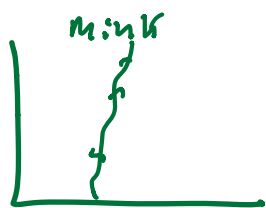
@ Equilibrium

QM $\xrightarrow{i\hbar \rightarrow \beta}$ Stat. Mech.

$$\sum_{\lambda} \langle \lambda | e^{-i\hat{H}t} | \lambda \rangle \rightarrow \sum_{\lambda} \langle \lambda | e^{-\beta \hat{H}} | \lambda \rangle = \text{Tr}[e^{-\beta \hat{H}}]$$

$$\Downarrow$$

$$\int D[\phi] e^{i \int_0^{t'} dt \int d^3x \mathcal{L}[\phi]} \xrightarrow[t \rightarrow i\tau]{t' \rightarrow -i\beta} \int_{\text{PBC}} D[\phi] e^{-\int_0^{\beta} d\tau \int d^3x \mathcal{L}[\phi]}$$



$$\phi(\tau=0) = \phi(\tau=\beta) \quad -\infty < \tau < \infty$$

$$\chi(x, \tau) = \sum_{n=-\infty}^{\infty} \chi_n(x) e^{i\omega_n \tau}$$

Where $\omega_n = \begin{cases} 2\pi n T & \text{Boson} \\ (2n+1)\pi T & \text{Fermion} \end{cases}$

Let's Implement this! ϕ^4 Example

$$V(\phi) = -\frac{\mu^2}{2} \phi^2 + \frac{\lambda}{4} \phi^4$$

$$\phi(x) = \phi_{cl} + \chi(x)$$

$\uparrow$ quantum correction

Zero temp 1-loop: $V'(\phi_{cl}) = \frac{1}{2(2\pi)^4} \int d^4k \ln[k^2 + 3\lambda\phi_{cl}^2 - \mu^2]$

Including PBC: $V'(\phi_{cl}, T) = \sum_n \frac{T}{2(2\pi)^3} \int d^3k \ln[(2\pi n T)^2 + k^2 + 3\lambda\phi_{cl}^2 - \mu^2]$

After renormalizing we find temp correct $m^2(\phi) = 3\lambda\phi_{cl}^2 - \mu^2$

$$V'_{\text{eff}}(\phi, T) = -\frac{\mu^2}{2} \phi^2 + \frac{\lambda}{4} \phi^4 + \frac{m^2(\phi)}{24} T^2 - \frac{\pi^2}{90} T^4$$

For electroweak (ignoring $V(1)$):

$$V'_{\text{eff}}(\phi, T) = D(T - T_0^2) \phi^2 - ET \phi^3 + \frac{\lambda}{4} \phi^4$$

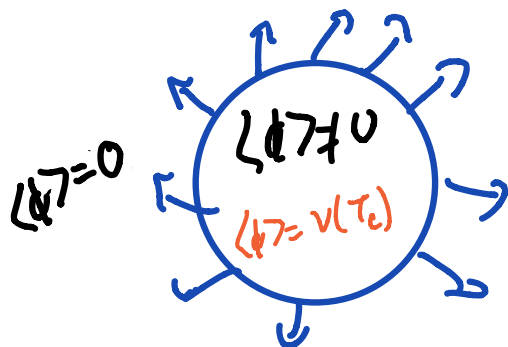
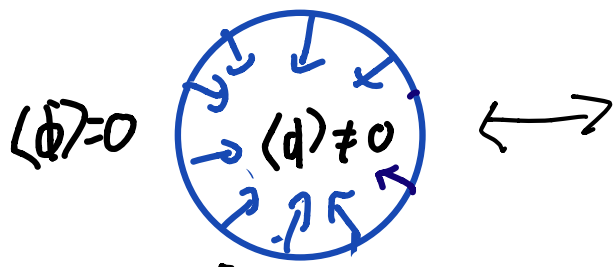
$$D = \frac{3}{32} g^2 + \frac{\lambda}{4} + \frac{m_t^2}{4v^2}$$

$$E = \frac{3g^2}{32\pi}$$

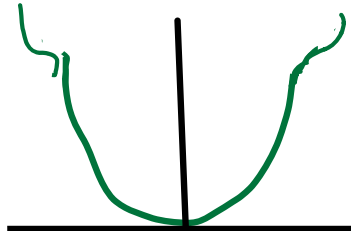
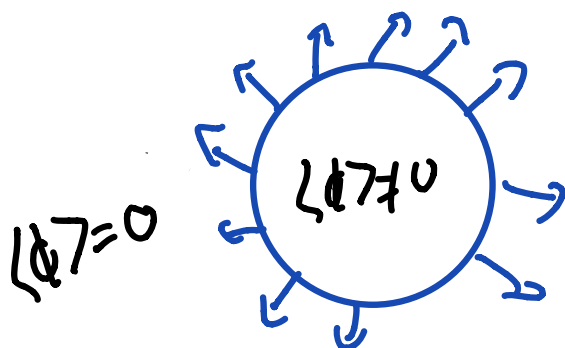
Must be strong 1st order

$$T_c = m_H \left(\frac{3}{8} g^2 + 1 - \frac{4}{256\pi^2} g^6 + \frac{m_t^2}{v^2} \right)^{-1/2}$$

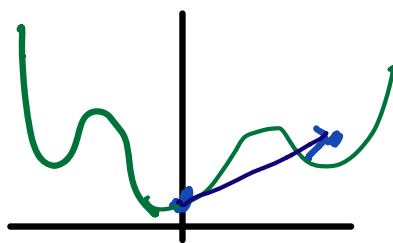
Bubble Shrink



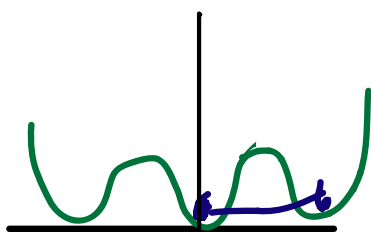
True Bubble Growth.



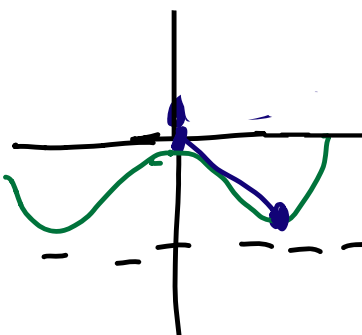
$$T \gg T_c$$



$$T > T_c$$



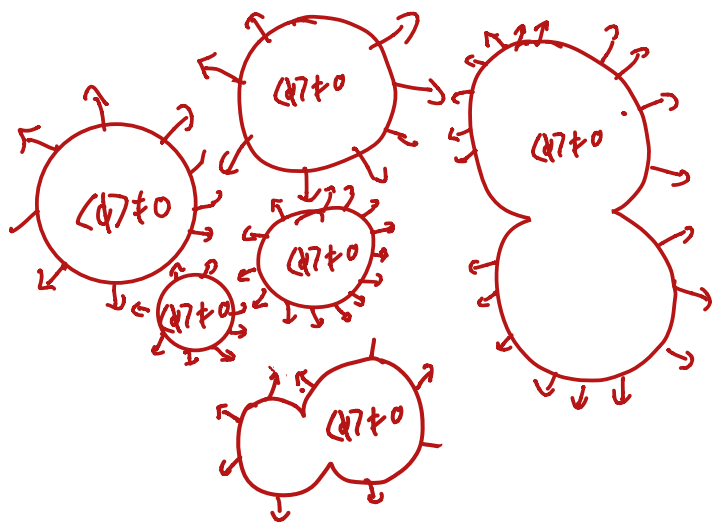
$$T = T_c$$



$$T \lesssim T_c$$

"Nucleation Temperature"
When nucleation rate exceeds that of hubble

Overview / Nucleation



Formula (simple)

Assume const. velocity v

$$\text{Volume: } V(t, t_0) = \frac{4\pi}{3} (t - t_0)^3 v^3$$

$$\Sigma(t) = \int_{t_0}^t dt_0 V(t, t_0) R(t_0)$$

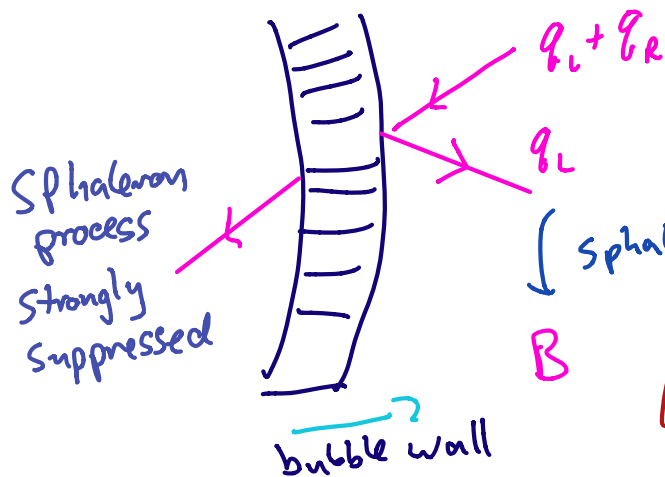
rate of nucleation

Fraction of total occupied space: $P = [1 - \exp(-\Sigma(t))]$

$$F_c[\phi, T] = \int dr r^2 \left[\frac{1}{2} (\phi')^2 + V(\phi, T) \right]$$

radius of nucleation: $R = \sqrt{\frac{34}{48\pi}} \quad (*)$

This is where EWBG takes place



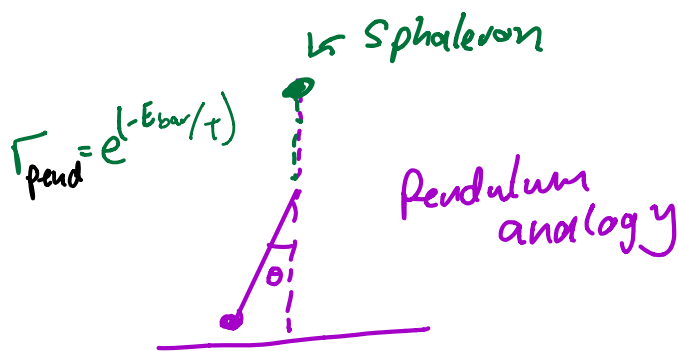
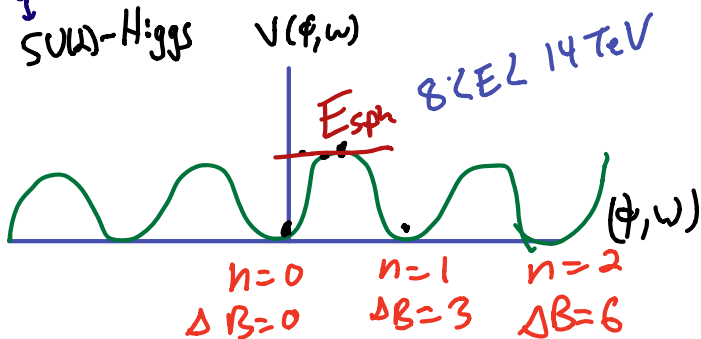
happens only in Unbroken Theory (Suppressed in broken)

Sphaleron process

B

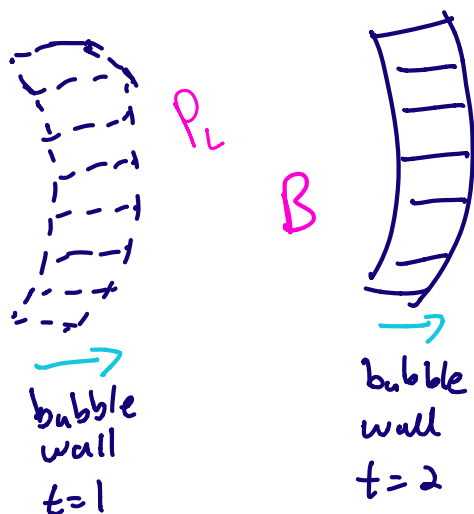
Whats a Sphaleron?

In full EW no analytic solution
SU(2)-Higgs



$$\Gamma_{tunnel}(\tau=0) \propto e^{-4\pi/g^2} \sim 10^{-170}$$

$$\Gamma_{thermal} \propto e^{[-E_{sph}/(kT)]} \quad T > 10^2$$



To Ensure No baryon production on the inside

$$\frac{E_{sph}}{T} \approx \frac{V(T)}{T_c} \gtrsim 1$$

This requirement implies the phase transition must be Strongly 1st order,

- New physics is needed.
- check $\rightarrow$ the MSSM is consistent w/ EWBG
- Problem can be solved coupling Scalars to Higgs.
- Axion couples weakly to Higgs

Anomalies $\implies$ Gauge Field Topology

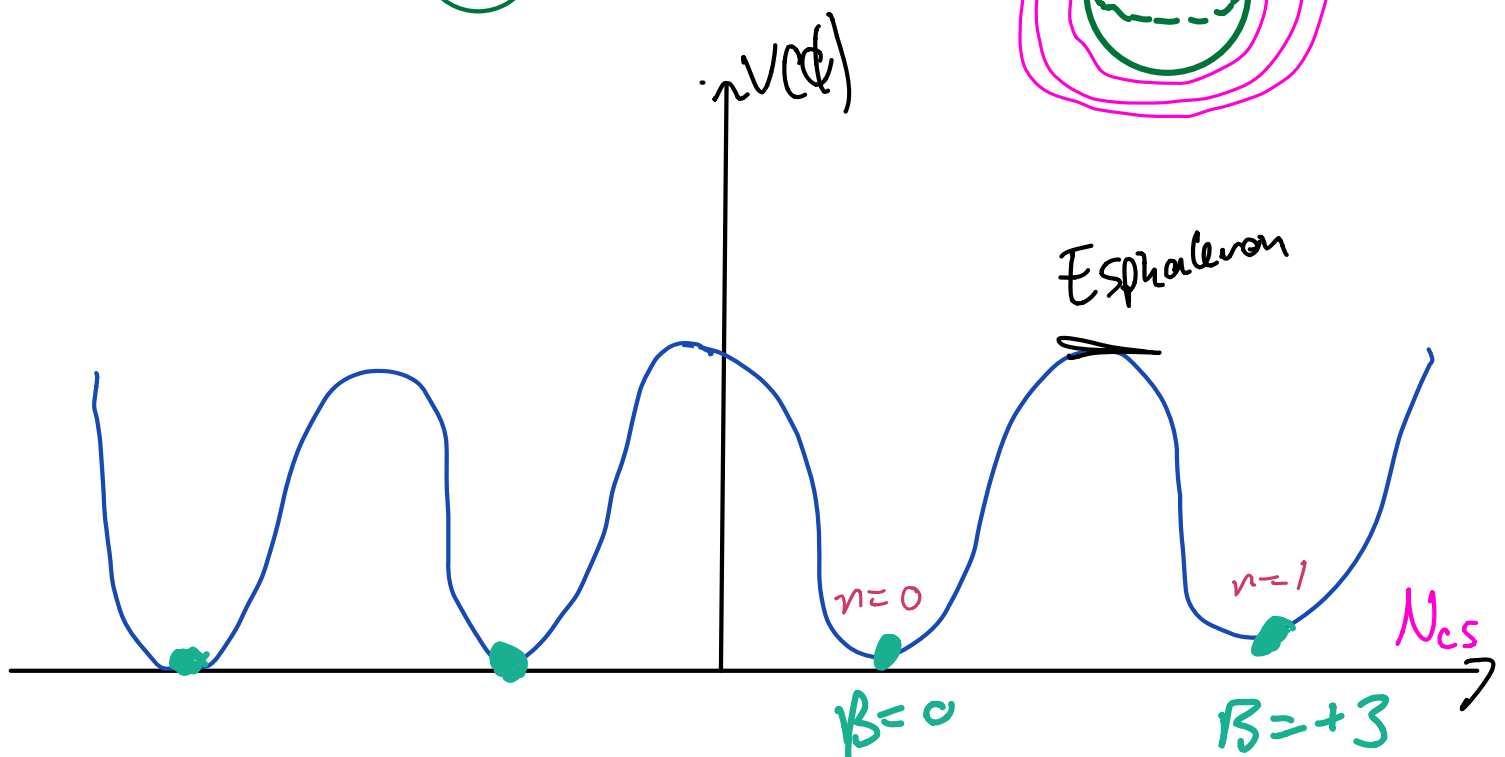
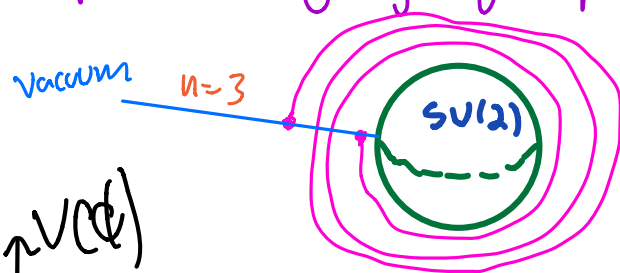
In class we showed the chiral anomalies:
(Adler-Bell-Jackiw)

$$2_n j^\mu = 2_n j^\mu = \eta_f \left(\frac{g^2}{32\pi^2} \overset{\text{Scr2}}{W}_{\mu\nu}^a \bar{W}^{\mu\nu a} + \frac{g'^2}{32\pi^2} \overset{U(1)}{F}_{\mu\nu} \tilde{F}^{\mu\nu} \right)$$

The **RHS** is related to the **Winding** gauge field

Gauge fields are maps from space to gauge group

$$SU(2): \mathbb{R}^3 \rightarrow \text{circle}(SU(2))$$



Chern-Simons # describes the winding

Yang Mills Lagrangian: $\mathcal{L} = -\frac{1}{2g^2} \text{Tr} F^{\mu\nu} F_{\mu\nu} + \frac{\theta}{16\pi^2} \underbrace{\text{Tr} [F^{\mu\nu} \tilde{F}_{\mu\nu}]}_{2 \, 2 \, \mu \, \nu \, \mu \, \nu}$

$$K^\mu = \frac{g^2}{32\pi^2} \epsilon^{\mu\nu\alpha\beta} \left(W_{\mu\nu}^a B_\beta^a - \frac{g}{3} \epsilon_{abc} B_\nu^a B_\alpha^b B_\beta^c \right)$$

$$2_{\mu} k^{\mu} = \frac{g^2}{32\pi^2} W_{\mu\nu}^a W^{\mu\nu a}$$

$$N_{CS} = \int d^3x k^0$$

N_{CS} is not gauge invariant, but:

$$N_{CS}(t_1) - N_{CS}(t_0) = \int_{t_0}^{t_1} dt \int d^3x 2_{\mu} k^{\mu} = \nu \in \mathbb{Z} \quad \text{for pure gauge}$$

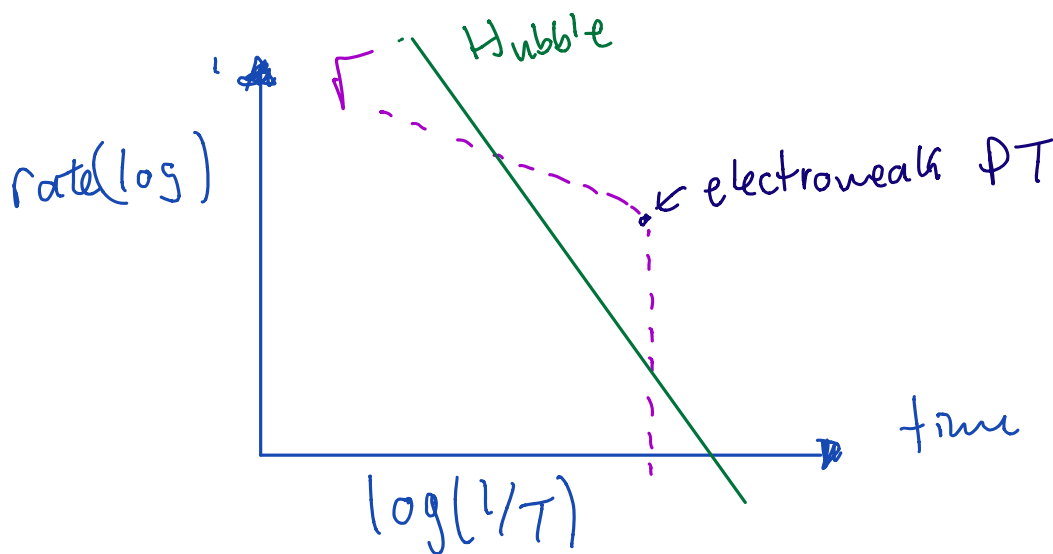
Ignoring U(1) $\approx k_{U(1)}^{\mu} = \frac{g'}{32\pi^2} \epsilon^{\mu\nu\alpha\beta} F_{\nu\alpha} B_{\beta}$

We find $2_{\mu} j_B^{\mu} = n_f 2_{\mu} k^{\mu}$!

$$\Delta B = 2_{\mu} j(t_f) - 2_{\mu} j(0) = n_f [2_{\mu} k^{\mu}(t_f) - 2_{\mu} k^{\mu}(0)] = n_f [N_{CS}(t_f) - N_{CS}(0)]$$

Recall: $A_{sphaleron} \sim e^{-E_{sph}/T}$ $A_{tunnel} \sim e^{-170} \Rightarrow 4S_E - \frac{32\pi^2}{g^2} N_{CS} > 0 \quad S_E \geq \frac{8\pi^2}{g^2} N_{CS}$

Sphaleron Process rate: $\frac{\Gamma}{V} = \text{const.} \left(\frac{E_{sph}}{T}\right)^3 \left(\frac{m_W(T)}{T}\right)^4 T^4 e^{-E_{sph}/T} = C \alpha_W^2 T^4$
 $\alpha_W = \frac{1}{30}$
 $E_{sph} = \frac{2m_W}{\alpha_W} f(\frac{1}{g^2})$
g-run by lattice computation



CP violations (Thin Wall)

occurs when complex phases that cannot be removed by field redefinition.

$$V_{CKM} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix} = \begin{pmatrix} c_1 & s_1 c_3 & s_1 s_3 \\ -s_1 c_2 & c_1 c_2 s_3 - s_2 s_3 e^{i\delta} & c_1 c_2 s_3 + s_2 s_3 e^{i\delta} \\ -s_1 s_2 & c_1 s_2 s_3 + c_2 s_3 e^{i\delta} & c_1 s_2 s_3 - c_2 s_3 e^{i\delta} \end{pmatrix}$$

not enough for CP violation w/o δ

To enhance CP violation

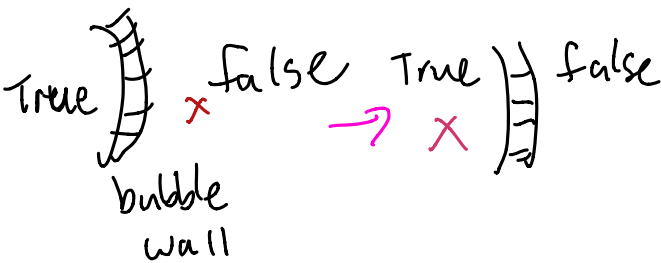
1.) 2 Higgs Doublet:

$$V(\phi_1, \phi_2, \theta) = \lambda_1 (\Phi_1^\dagger \Phi_1 - v_1^2)^2 + \lambda_2 (\Phi_2^\dagger \Phi_2 - v_2^2)^2 + \lambda_3 [(\Phi_1^\dagger \Phi_1 - v_1^2) + (\Phi_2^\dagger \Phi_2 - v_2^2)]^2 + \lambda_4 [(\Phi_1^\dagger \Phi_1)(\Phi_2^\dagger \Phi_2) - (\Phi_1^\dagger \Phi_2)(\Phi_2^\dagger \Phi_1)] + \lambda_5 [\text{Re}(\Phi_1^\dagger \Phi_2) - v_1 v_2 \cos \theta]^2 + \lambda_6 [\text{Im}(\Phi_1^\dagger \Phi_2) - v_1 v_2 \sin \theta]^2$$

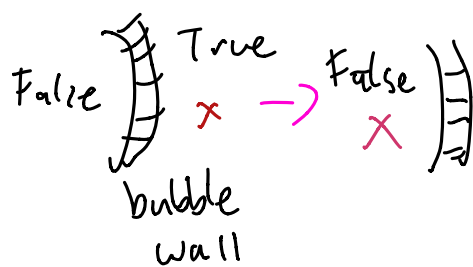
In Unitary gauge: $\Phi_1 = (0, \phi_1)^T$ $\Phi_2 = (0, \phi_2 e^{i\theta})^T$ Invariant phase cannot be removed by field redef

Free Energy can be calculated: $F = - \frac{14}{3\pi^2 n_f} \zeta(3) \left(\frac{m}{T}\right)^2 \dot{\theta} n_B = \mu_B n_B$ mass to particle species

μ_B



$\Delta\theta > 0$ Baryon Production



$\Delta\theta < 0$ anti-Baryon Production

$$\Delta\theta \sim \xi - \arctan\left(\frac{16}{\lambda_5} \tan \zeta\right)$$

Punchline: $\frac{dn_B}{dt} = -3 \frac{\Gamma(T)}{T} \Delta F$

Ways to test EWBG

Needs new physics Strong first order and Provide CP violation

For CPV we look at EDM

Puts constraints on new particles

$$|\sin \phi| \leq (M/2 \text{ TeV})^2$$

But this can be evaded with new physics.

Can create new particles at LHC

lighter than $M_x \lesssim 200 \text{ GeV}$ testable constraint

Strong first-order phase transition can be produced by gravity waves.

Gravitational radiation is formed by turbulent expansion of bubbles

Depends on

- latent heat released by PT at T_n relative to background radiation energy.

- Rate of bubble nucleation β

- Bubble wall velocity

MSSM provides naturally sufficient CP violation

Stops coupling to higgs provide a strong first-order PT

For CP violation we can introduce higher dim operators